

PROGRAM 1 (TIC-TAC-TOE)

TIC-TAC-TOE USING MINIMAX AI

Board representation

```
board = [' ' for _ in range(9)]
```

HUMAN = 'X'

COMPUTER = 'O'

Display the board

```
def display_board():
```

```
    print()
```

```
    print(f" {board[0]} | {board[1]} | {board[2]} ")
```

```
    print("----+----+---")
```

```
    print(f" {board[3]} | {board[4]} | {board[5]} ")
```

```
    print("----+----+---")
```

```
    print(f" {board[6]} | {board[7]} | {board[8]} ")
```

```
    print()
```

Check whether a player has won

```
def check_win(player):
```

```
    winning_combinations = [
```

```
        (0, 1, 2), # Row 1
```

```
        (3, 4, 5), # Row 2
```

```
        (6, 7, 8), # Row 3
```

```
        (0, 3, 6), # Column 1
```

```
        (1, 4, 7), # Column 2
```

```
        (2, 5, 8), # Column 3
```

```
(0, 4, 8), # Main diagonal
(2, 4, 6) # Other diagonal
]
```

```
for a, b, c in winning_combinations:
```

```
    if board[a] == player and board[b] == player and board[c] == player:
```

```
        return True
```

```
return False
```

```
# Check whether the board is full
```

```
def check_draw():
```

```
    return ' ' not in board
```

```
# Minimax algorithm
```

```
def minimax(is_maximizing):
```

```
    # Terminal states
```

```
    if check_win(COMPUTER):
```

```
        return 10
```

```
    if check_win(HUMAN):
```

```
        return -10
```

```
    if check_draw():
```

```
        return 0
```

```
# Computer's turn - maximize score
```

```
if is_maximizing:

    best_score = -float('inf')

    for i in range(9):

        if board[i] == ' ':

            board[i] = COMPUTER

            score = minimax(False)

            board[i] = ' '

            best_score = max(best_score, score)

    return best_score

# Human's turn - minimize score
else:

    best_score = float('inf')

    for i in range(9):

        if board[i] == ' ':

            board[i] = HUMAN

            score = minimax(True)

            board[i] = ' '

            best_score = min(best_score, score)
```

```
    return best_score

# Find the best move for the computer
def find_best_move():
    best_score = -float('inf')
    best_move = None

    for i in range(9):
        if board[i] == ' ':
            board[i] = COMPUTER

            score = minimax(False)

            board[i] = ' '

            if score > best_score:
                best_score = score
                best_move = i

    return best_move

# Human move
def human_move():
    while True:
        try:
            position = int(input("Enter your position (1-9): "))
```

```

    if position < 1 or position > 9:

        print("Please enter a number from 1 to 9.")

    elif board[position - 1] != ' ':

        print("That position is already occupied.")

    else:

        board[position - 1] = HUMAN

        break

except ValueError:

    print("Please enter a valid number.")

# Main game

def play_game():

    global board

    board = [' ' for _ in range(9)]

    print("=====")
    print("    TIC-TAC-TOE GAME")
    print("=====")
    print("You are X")
    print("Computer is O")
    print()
    print("Board positions are:")
    print()
    print(" 1 | 2 | 3 ")
    print("----+---+---")

```

```
print(" 4 | 5 | 6 ")
```

```
print("----+----+---")
```

```
print(" 7 | 8 | 9 ")
```

```
print("\nGame starts!\n")
```

```
while True:
```

```
    # Human's turn
```

```
    display_board()
```

```
    print("Your turn.")
```

```
    human_move()
```

```
    # Check human win
```

```
    if check_win(HUMAN):
```

```
        display_board()
```

```
        print("Congratulations! You WIN!")
```

```
        break
```

```
    # Check draw
```

```
    if check_draw():
```

```
        display_board()
```

```
        print("The game is a DRAW!")
```

```
        break
```

```
    # Computer's turn
```

```
    print("Computer is thinking...")
```

```
best_move = find_best_move()
board[best_move] = COMPUTER
```

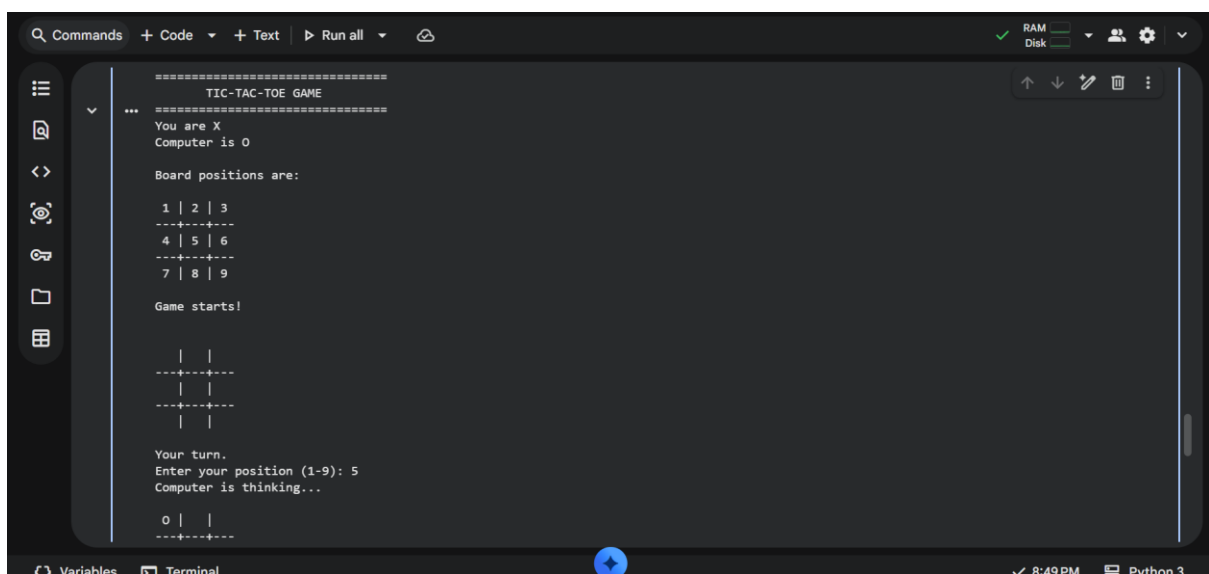
```
# Check computer win
if check_win(COMPUTER):
    display_board()
    print("Computer WINS!")
    break
```

```
# Check draw
if check_draw():
    display_board()
    print("The game is a DRAW!")
    break
```

```
# Start the game
```

```
play_game()
```

OUTPUT



```
=====
TIC-TAC-TOE GAME
=====
...
You are X
Computer is O

Board positions are:

 1 | 2 | 3
---+---+---
 4 | 5 | 6
---+---+---
 7 | 8 | 9

Game starts!

  |  |
---+---+---
  |  |
---+---+---
  |  |

Your turn.
Enter your position (1-9): 5
Computer is thinking...

O |  |
---+---+---
```

Q Commands+ Code+ TextRun all

RAMDisk

Enter your position (1-9): 5

Computer is thinking...

...

o | |

| x |

| |

Your turn.

Enter your position (1-9): 3

Computer is thinking...

o | | x

| x |

o | |

Your turn.

Enter your position (1-9): 2

Computer is thinking...

o | x | x

o | x |

o | |

Your turn.

Enter your position (1-9): 2

Computer is thinking...

o | x | x

o | x |

o | |

Computer WINS!

+ Code

+ Text